Button

DETECT | PICO | INPUT

What does it do?

A button lets someone tell your program to do something the moment they press.

Inside are two metal contacts. Pressing pushes them together so electricity can flow, and letting go springs them apart again.

Your projects can use it to start something, stop something, or count how many times it has been pressed.

Uses



Pedestrian crossing

DETECT: Button and traffic phase → DECIDE: Is it safe to cross? → DO: Change the lights



Do not press this

DETECT: Button is pressed → DECIDE: Is the trap armed? → DO: Play a silly sound



Cap
Press here

Legs
Four, joined in pairs

Gap
Straddle the channel

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Button

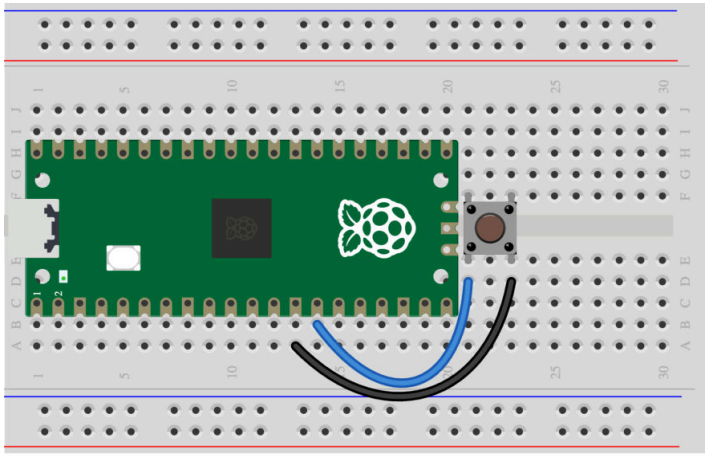
button = Button(10)

pressed = button.is_pressed
```

`pressed` is True or False. No resistor is needed — picozero switches on the pin's own pull-up. Also `button.when_pressed` and `button.when_released`.

Wiring

PICO



GP10 (pin 14) → button → **GND** (pin 13)

Capacitive touch

DETECT PICO INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses



Touch lamp

DETECT
Finger touches the pad
→
DECIDE
Is the lamp already on?
→
DO
Toggle the light



Banana piano

DETECT
A banana is touched
→
DECIDE
Which note should play?
→
DO
Play the note



Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import TouchSensor

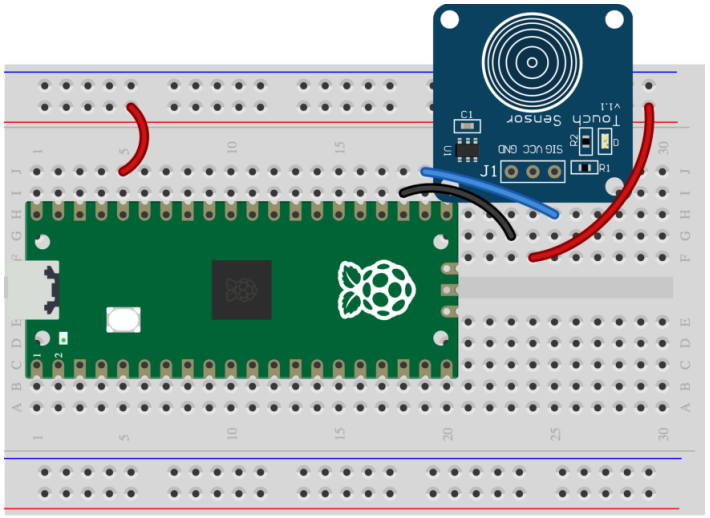
touch = TouchSensor(17)

touched = touch.is_touched
```

The TTP223 output goes high while touched. `touched` is True or False.

Wiring

PICO



GND (pin 23) → GND, **3V3 OUT** (pin 36) → VCC, **GP17** (pin 22) → SIG

LDR

DETECT PICO LIGHT

What does it do?

An LDR is a sensor that measures how bright or dark it is.

LDR stands for light-dependent resistor. Light shining on it lets electricity through more easily, so your code can tell light from dark.

Your projects can use it to react when a lamp comes on, when night falls, or when a shadow passes over.

Uses



Street light

DETECT How dark it is → **DECIDE** Is it darker than dusk? → **DO** Switch the lamp on



Biscuit tin alarm

DETECT Light inside the tin → **DECIDE** Is the lid open? → **DO** Sound the alarm



Window Light lands here

Track Zigzag sensor

Legs Either way round

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import AnalogInputDevice

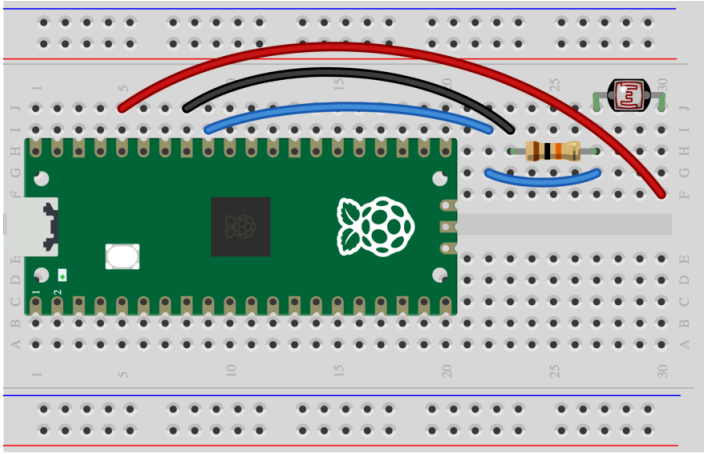
ldr = AnalogInputDevice(27)

light = ldr.value
```

`light` runs from 0 in the dark to 1 in bright light.
Also `ldr.voltage`.

Wiring

PICO



3V3 (pin 36) → LDR → **GP27** (pin 32) → 10 kΩ → **GND** (pin 33)

Line sensor

DETECT PICO INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

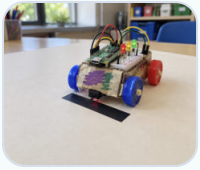
Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



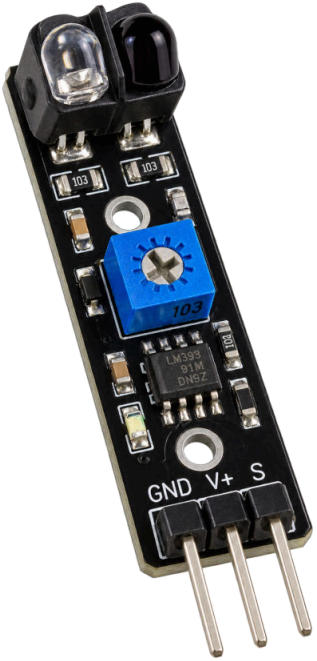
Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import DigitalInputDevice

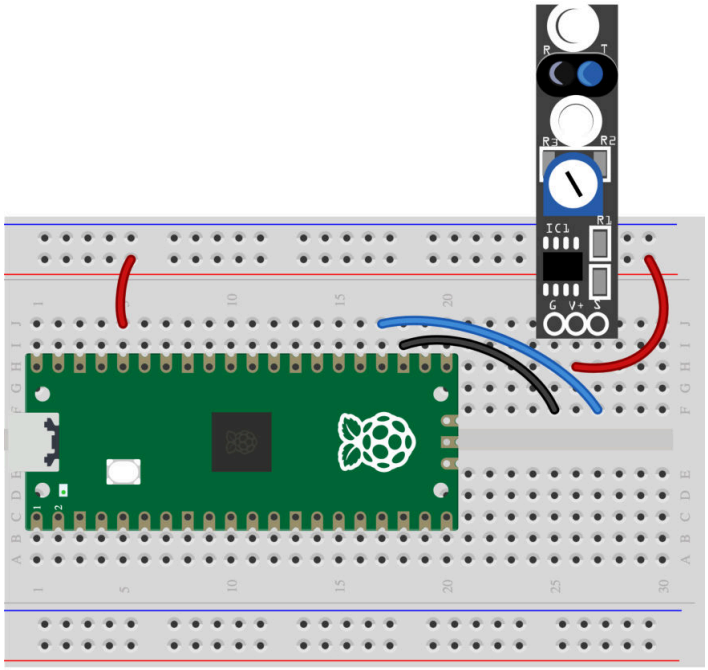
line = DigitalInputDevice(18, pull_up=False)

on_line = line.is_active
```

Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

Wiring

PICO



GND (pin 23) → G, **3V3 OUT** (pin 36) → V+, **GP18** (pin 24) → S

Distance sensor

DETECT PICO DISTANCE

What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

Uses



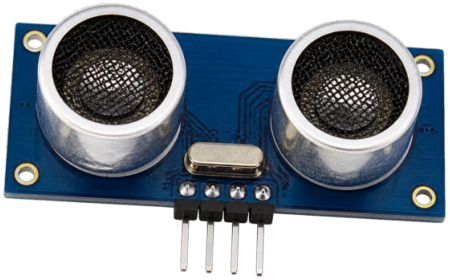
Parking sensor

DETECT Gap to the wall → **DECIDE** Is it closer than 30 cm? → **DO** Beep faster



Air harp

DETECT How far your hand is → **DECIDE** Which note is that? → **DO** Play the note



VCC	Trig	Echo	GND
Power, 3V3	Sends a click	Times the return	Ground

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import DistanceSensor

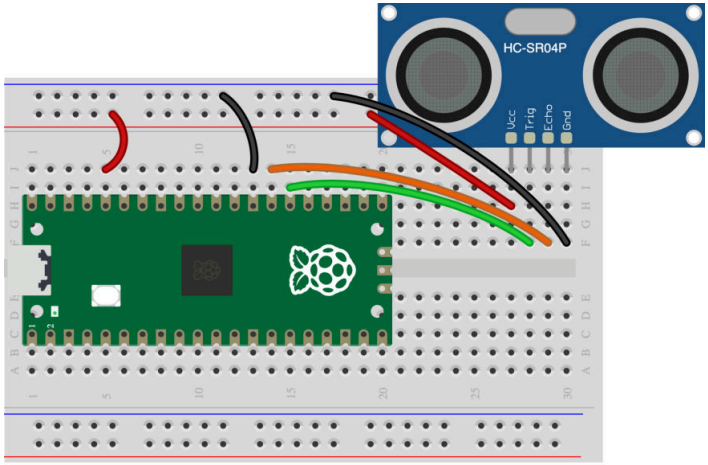
sensor = DistanceSensor(
    echo=21, trigger=20
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Use an **HC-SR04P**, powered from **3V3**.

Wiring

PICO



HC-SR04P · Trig **GP20** (pin 26) · Echo **GP21** (pin 27) · **3V3 OUT** (pin 36) to top + rail · **GND** (pin 28) to top − rail

PIR

DETECT PICO MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

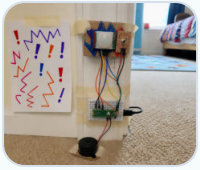
Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses



Burglar alarm

DETECT Movement in the room → **DECIDE** Should the room be empty? → **DO** Raise the alarm



Sibling alarm

DETECT Movement at the door → **DECIDE** Is privacy mode armed? → **DO** Sound the warning



Dome Wide-angle lens

VCC Power, 5V

OUT High on movement

GND Ground

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import MotionSensor

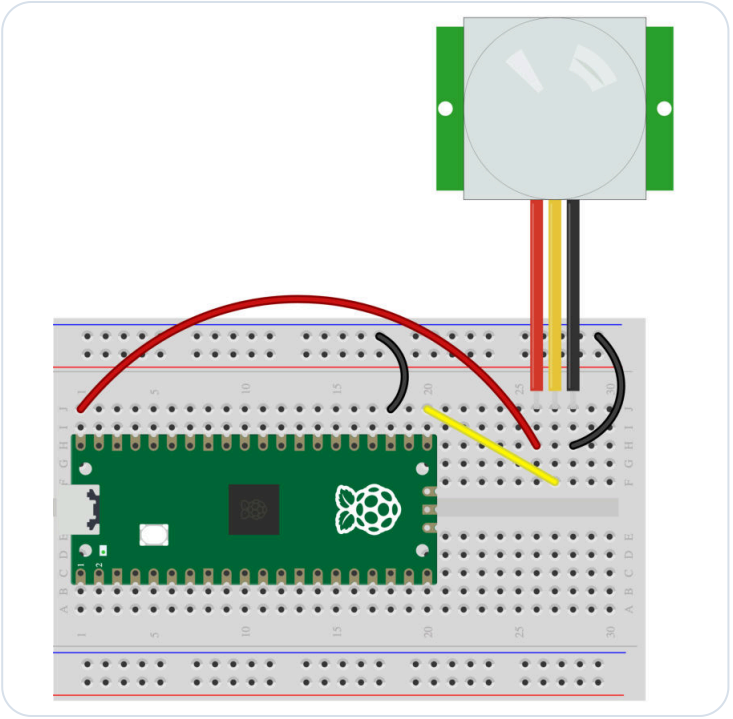
pir = MotionSensor(16)

moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

Wiring

PICO



VBUS (pin 40) · OUT **GP16** (pin 21) · **GND** (pin 23)

AM312 PIR

DETECT PICO MOTION MINI PIR

What does it do?

An **AM312** is a tiny PIR sensor that notices warm bodies moving nearby.

PIR means passive infrared. Its faceted lens divides the view into zones, so a person or animal moving between them makes the output switch on.

The complete AM312 module runs from 3V3 and uses very little power. Its sensitivity and roughly two-second output pulse are fixed.

Uses



Wardrobe light

DETECT Movement in the wardrobe → **DECIDE** Is someone reaching in? → **DO** Turn on the light



Dragon treasure alarm

DETECT Movement near the hoard → **DECIDE** Is someone sneaking closer? → **DO** Make the dragon roar



Lens	VCC	OUT	GND
Splits the view into zones	Power from 3V3	High on movement	Ground

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import MotionSensor

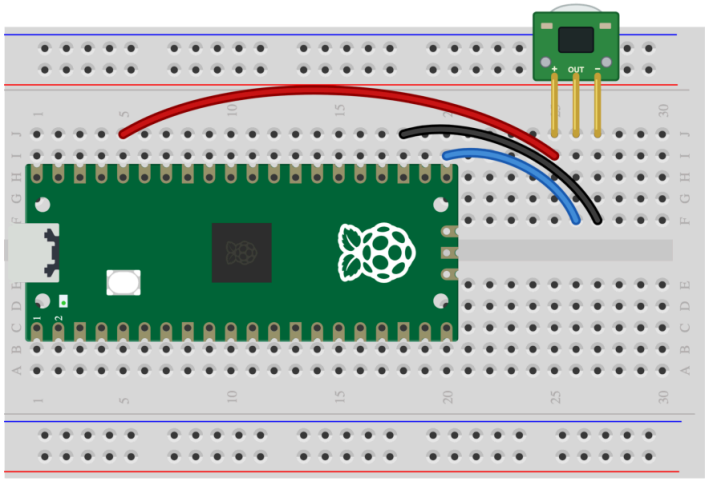
pir = MotionSensor(16)

moved = pir.motion_detected
```

`moved` is True or False. Use `pir.when_motion` and `pir.when_no_motion` to run functions when it changes. The AM312 has no sensitivity or delay dials; OUT normally stays high for about two seconds after movement.

Wiring

PICO



VCC → 3V3 OUT (pin 36); **OUT** → GP16 (pin 21); **GND** → GND (pin 23).

Potentiometer

DETECT PICO INPUT

What does it do?

A potentiometer lets your code read how far a knob has been turned.

This small version is called a trimpot. Turning the blue knob moves a contact called the wiper, changing the voltage your board reads.

Use it to choose a speed, a brightness or a sound. Turn gently and stop at the ends. The marking 103 means 10 kΩ.

Uses



Security-light adjustment

DETECT Read sensitivity setting → DECIDE Is movement strong enough? → DO Switch the light on



Alien voice tuner

DETECT Knob position → DECIDE Should it rumble or squeak? → DO Rumble or squeak



Knob	Outer pair	Wiper
Turn to adjust	Pins 1 and 3	Pin 2, offset

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Pot

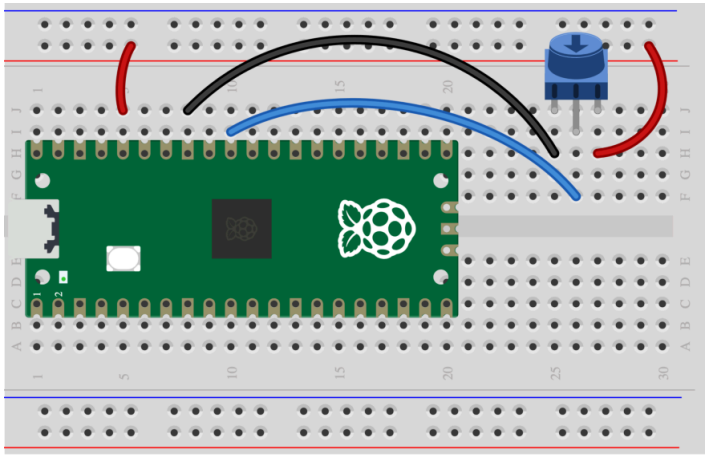
pot = Pot(26)

setting = pot.value
```

`setting` runs from 0 to 1. Read `pot.value` again after turning; `pot.voltage` gives volts. Power the trimpot from **3V3**.

Wiring

PICO



Trimpot pin **1** → **GND** (pin 33) · **2** (wiper) → **GP26** (pin 31) · **3** → **3V3 OUT** (pin 36). Each pin has its own breadboard row.

Temperature sensor

DETECT | PICO | INPUT | ANALOGUE

What does it do?

A TMP36 turns temperature into a changing voltage that a program can measure.

Its output rises by 10 millivolts for every degree Celsius, with 0.5 V representing 0 °C.

The Pico reads that voltage directly; the Raspberry Pi uses an MCP3008 analogue-to-digital converter.

Uses



Refrigerator cooling

DETECT Measure fridge temperature → **DECIDE** Is it too warm? → **DO** Increase the cooling



Soup drama meter

DETECT Measure bowl surface → **DECIDE** Which drama level fits? → **DO** Move the drama pointer



Flat face	+Vs	Vout	GND
Use it to identify the pins	Left pin	Centre pin	Right pin

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import TemperatureSensor

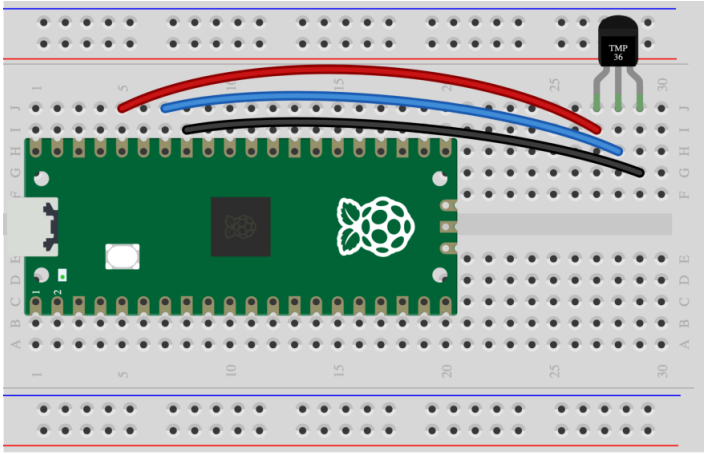
convert = lambda voltage: (voltage - 0.5) * 100
sensor = TemperatureSensor(28, conversion=convert)

temperature = sensor.temp
```

Read the TMP36 with its flat face towards you; the pins are +Vs, Vout, GND from left to right.

Wiring

PICO



TMP36 **+Vs** → 3V3 OUT (pin 36), **Vout** → GP28 (pin 34), **GND** → GND (pin 33)

Temperature probe

DETECT | PICO | DS18B20 | DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses



Aquarium temperature alarm

DETECT Measure tank temperature → **DECIDE** Is it too cold? → **DO** Show a warning



Penguin bath alarm

DETECT Measure bath temperature → **DECIDE** Is it too warm? → **DO** Light a warning LED



VDD	DQ	GND	4.7 kΩ
Power from 3V3	Digital data	Ground	Resistor from DQ to 3V3

Raspberry Pi Foundation

Code

Raspberry Pi Pico

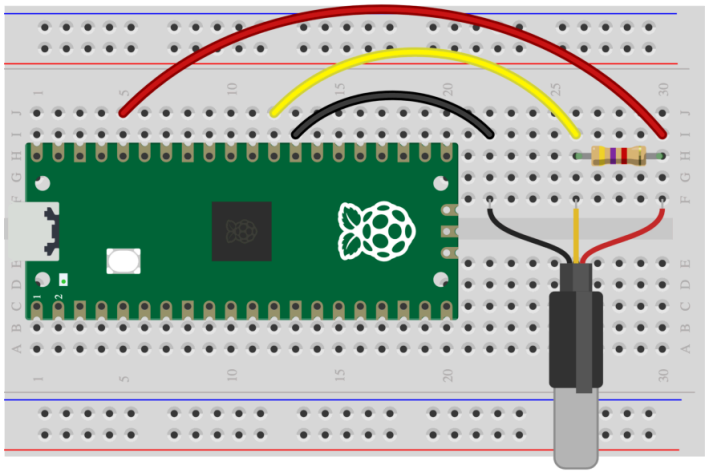
```
from machine import Pin
import onewire, ds18x20
from time import sleep_ms

bus = onewire.OneWire(Pin(22))
sensor = ds18x20.DS18X20(bus)
probes = sensor.scan()
assert probes, "Check probe wiring"
sensor.convert_temp()
sleep_ms(750)
temperature = sensor.read_temp(probes[0])
print(temperature)
```

Repeat the conversion and 750 ms wait for each new reading.

Wiring

PICO



VDD → 3V3 OUT (pin 36); **DQ** → GP22 (pin 29); **GND** → GND (pin 28). **4.7 kΩ** between DQ and 3V3.

Tilt switch

DETECT PICO INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

While it stands upright, a tiny metal ball rests on two contacts and completes the circuit. Tip it over and the ball rolls away, breaking the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Iron safety shut-off

DETECT: Iron tips sideways → DECIDE: Has the delay elapsed? → DO: Switch the heater off



Grumpy pencil case

DETECT: Case is tipped → DECIDE: Has it stayed tipped? → DO: Make a grumpy noise



Metal can
Contains the rolling ball

Legs
No polarity

Angle
Changes the contact

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Switch

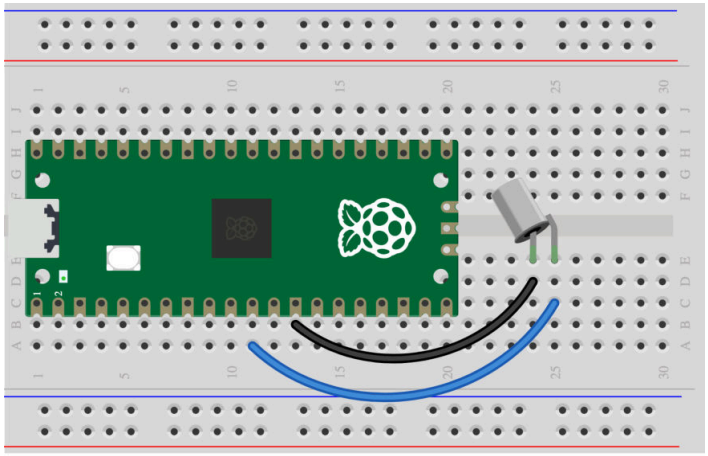
tilt = Switch(8)

tilted = tilt.is_open
```

> The reading may flicker while the ball rolls. Check it more than once before treating it as a definite tilt.

Wiring

PICO



GP8 (pin 11) → tilt switch → **GND** (pin 13)

Raspberry Pi Foundation

Toggle switch

DETECT | PICO | INPUT

What does it do?

A toggle switch stays in the position you move it to, unlike a push button.

Its centre COM leg connects to one outside leg in each position.

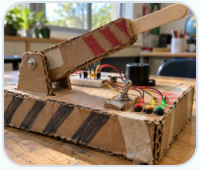
It is useful when a project needs a lasting on/off choice or two modes.

Uses



Auto/manual selector

DETECT Read selector position → **DECIDE** Is automatic mode selected? → **DO** Run the selected mode



Serious business lever

DETECT Lever and check buttons → **DECIDE** Are all checks complete? → **DO** Start unnecessary sirens



Lever Stays where it is moved

COM Centre pin

L1 and L2 The two choices

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Switch

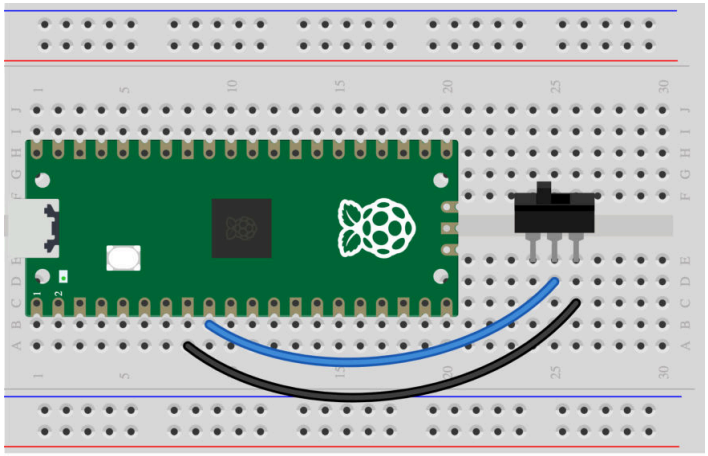
mode = Switch(6)

mode_two = mode.is_closed
```

Connect COM and one outside pin. The unused outside pin can be left unconnected.

Wiring

PICO



GP6 (pin 9) → COM; right-hand outside pin → **GND** (pin 8)

LED

DO PICO LIGHT

What does it do?

An LED is a tiny light that your code can switch on, switch off, or blink.

LED stands for light-emitting diode. It glows when electricity flows the right way through it, so the long leg and the short leg matter.

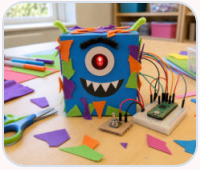
Your projects can use LEDs to show a warning, celebrate something, or tell people what is happening.

Uses



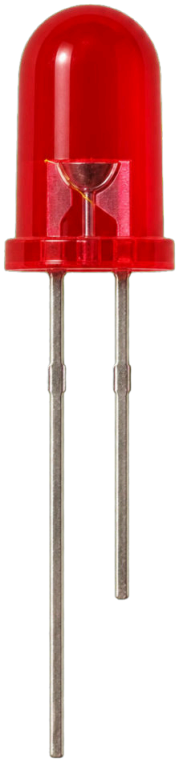
Car brake light

DETECT Brake control signal → **DECIDE** Is braking requested? → **DO** Light the red LEDs



One-eyed monster box

DETECT Box lid is opened → **DECIDE** Is guard mode armed? → **DO** Glow the monster eye



Anode
Long leg, to GPIO

Cathode
Short leg, to GND

Lens
Glows here

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import LED

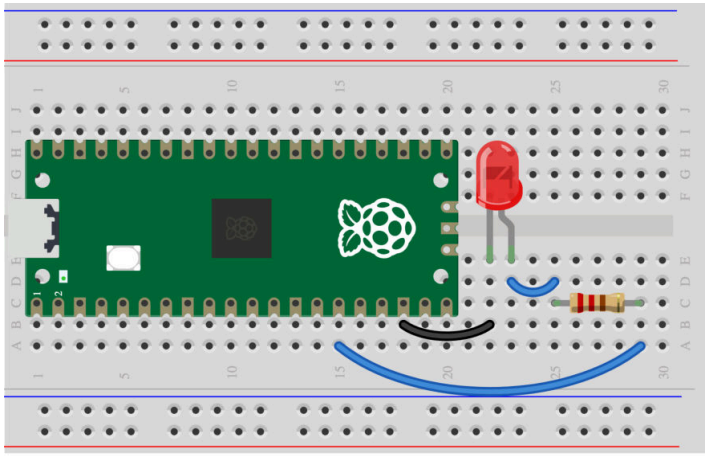
led = LED(11)

led.on()
```

Also `led.off()`, `led.toggle()` and `led.blink()`. Give `blink()` an `on_time` and an `off_time` to set the flash speed.

Wiring

PICO



GP11 (pin 15) → 220 Ω → LED → **GND** (pin 18)

Raspberry Pi Foundation

RGB LED

DO PICO OUTPUT LIGHT

What does it do?

An RGB LED puts red, green and blue lights inside one clear package.

Varying the brightness of the three colours mixes millions of possible colours.

A common-cathode LED has one shared negative leg and needs one resistor for each colour.

Uses



Wi-Fi router status

DETECT Read connection status → **DECIDE** Which status colour is needed? → **DO** Mix the LED colour



Mood crystal

DETECT Read the mood dial → **DECIDE** What is the crystal's mood? → **DO** Glow that colour



Clear lens
Contains three LEDs

Long leg
Common cathode

R, G and B legs
One 220 ohm resistor each

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import RGBLED

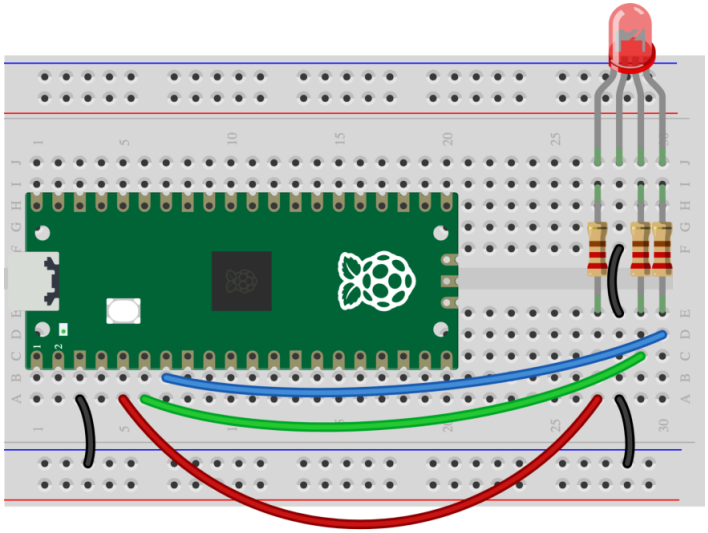
light = RGBLED(red=3, green=4, blue=5)

light.color = (255, 0, 255)
```

picozero colour values run from 0 to 255. This example makes magenta.

Wiring

PICO



R → **GP3** (pin 5), G → **GP4** (pin 6), B → **GP5** (pin 7); common → **GND** (pin 3)

NeoPixels

DO | PICO | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels



Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels



Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

VDD and GND 3.3 V power

Raspberry Pi Foundation

Code

Raspberry Pi Pico

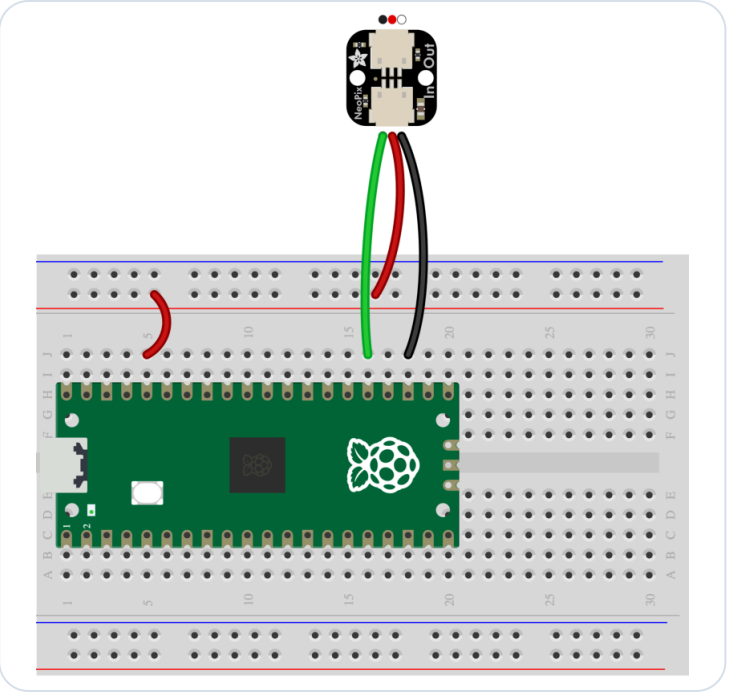
```
from machine import Pin
from neopixel import NeoPixel

pixels = NeoPixel(Pin(19), 1)
pixels[0] = (255, 0, 80)
pixels.write()
```

This wiring is for the pictured 3.3 V-compatible breakout. Generic 5 V strips usually need a separate supply and a logic-level shifter.

Wiring

PICO



GP19 (pin 25) → IN, **3V3 OUT** (pin 36) → top + rail → VDD, **GND** (pin 23) → GND

Active buzzer

DO PICO SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

Use a labelled **3.3 V GPIO-compatible** buzzer.

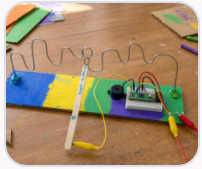
Uses



Smoke alarm

One warning tone is enough; no changing notes needed.

DETECT Measure smoke level → **DECIDE** Is smoke above the limit? → **DO** Repeat one warning tone



Buzz wire game

Choose active: one fixed buzz clearly signals failure.

DETECT Loop touches wire → **DECIDE** Is this the third strike? → **DO** Buzz at one pitch



+ leg Marked +, to GPIO

– leg Other leg, to GND

Vent Sound comes out

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Buzzer

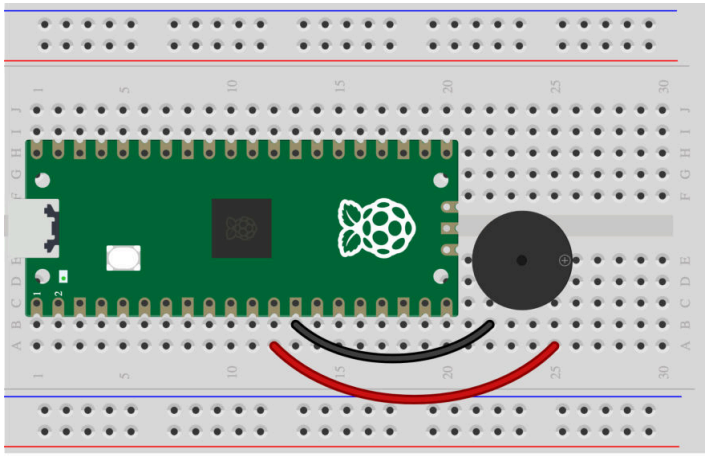
buzzer = Buzzer(9)

buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm. Faster beeps do not mean a higher pitch.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **–** → **GND** (pin 13)

Passive buzzer

DO PICO SOUND PWM

What does it do?

A passive buzzer needs fast pulses (PWM) to make sound. Changing their frequency changes its pitch: low or high notes.

Recognise it: check for **passive**, **external drive** or **no oscillator** on the packet. Both types can look identical; + only marks polarity.

Use a small **3.3 V GPIO-compatible piezo** sounder.

Uses



Washing-machine chime

PWM plays tunes; active buzzers have only one pitch.

DETECT Read cycle status → **DECIDE** Has the wash finished? → **DO** Play a short tune



Robot voice box

Change pitch for happy chirps or grumpy growls.

DETECT Read mood button → **DECIDE** Is happy pressed? → **DO** Chirp high or low



+ leg
Marked +, to GPIO

− leg
Other leg, to GND

Vent
Sound comes out

Raspberry Pi Foundation

Code

rpf.io/soundfx

Raspberry Pi Pico

```
from picozero import PWMBuzzer

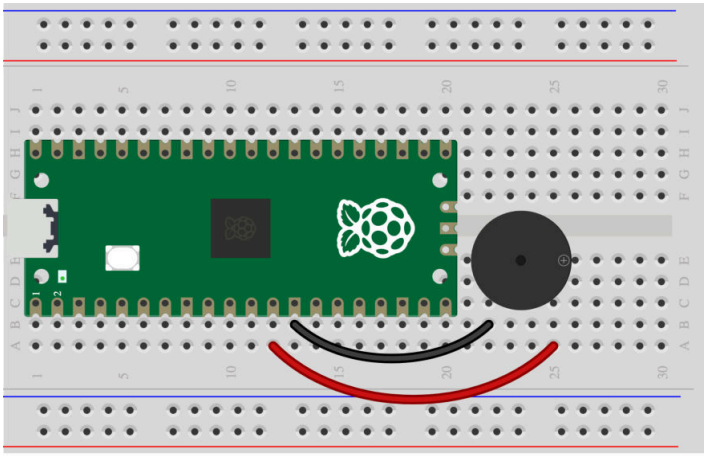
# GP9, 2000 Hz, 50% duty
buzzer = PWMBuzzer(9, 2000, 32768)

buzzer.on()
```

Try `buzzer.freq = 1000`; silence with `buzzer.off()`. Higher frequency means higher pitch. Beep speed is separate; both types can make rhythms.

Wiring

PICO



GP9 (pin 12) → **+** buzzer **–** → **GND** (pin 13)

Speaker

DO | PICO | OUTPUT | SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes



Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes



Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Speaker

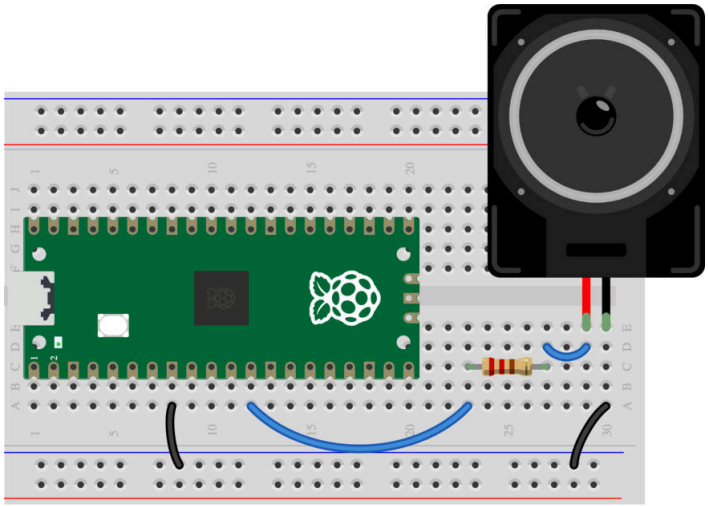
speaker = Speaker(9)

speaker.play("E4", duration=1)
```

> For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

Wiring

PICO



GP9 (pin 12) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 8) via the − rail

Motor driver

DO | PICO | OUTPUT | MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

The L293D contains H-bridges that can send current through a motor in either direction.

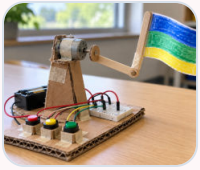
Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses



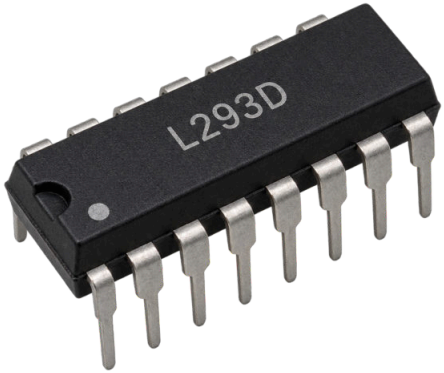
Electric car window

DETECT Read window switch → **DECIDE** Can it move safely? → **DO** Drive requested direction



Tiny flag waver

DETECT Count score-button presses → **DECIDE** Has the target been reached? → **DO** Wave the flag



Notch
Identifies pin 1

IN1 and IN2
Direction controls

OUT1 and OUT2
Connect to the motor

Enable
Held high in this circuit

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Motor

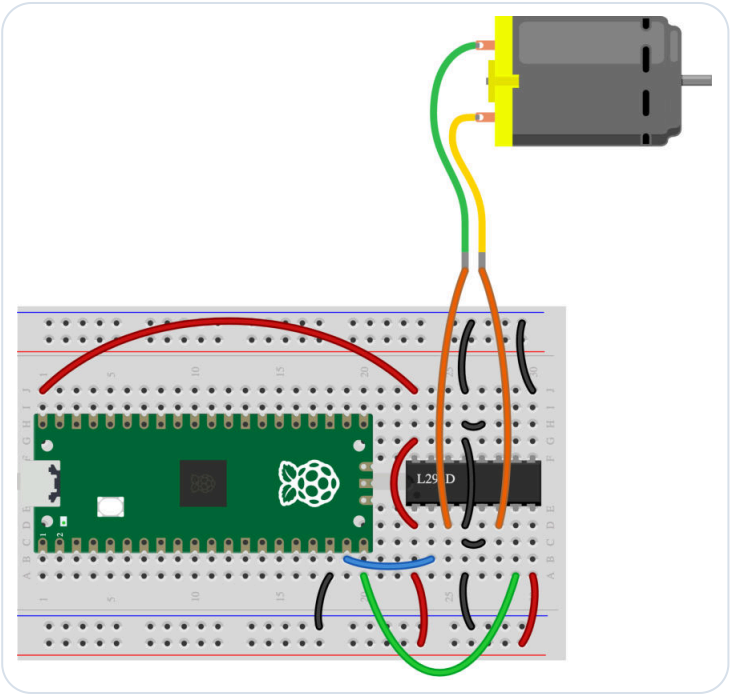
motor = Motor(14, 15)

motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

Wiring

PICO



GP14 (pin 19) → IN1, **GP15** (pin 20) → IN2; OUT1 and OUT2 → motor; **VBUS** (pin 40), **GND** (pin 18)

Servo motor

DO | PICO | OUTPUT | MOVEMENT

What does it do?

A positional servo moves its arm, called a horn, to a chosen position and holds it there.

Short pulses on its signal wire set the position. The motor and position control are built into the case.

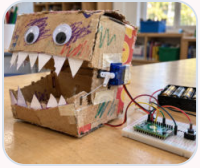
Use it for steering, pointing or opening a flap. A continuous-rotation servo controls speed instead; it cannot use this example to choose an angle.

Uses



Radio-controlled steering

DETECT Read steering command → **DECIDE** Which way should it turn? → **DO** Set the steering position



Cardboard chomping monster

DETECT Read feeding button → **DECIDE** Is the monster hungry? → **DO** Open its cardboard jaw



Horn	Signal	Power	Ground
Attach a light moving part	Usually orange, yellow or white	Usually red; about 5 V	Brown or black; join grounds

Raspberry Pi Foundation

Code

Raspberry Pi Pico

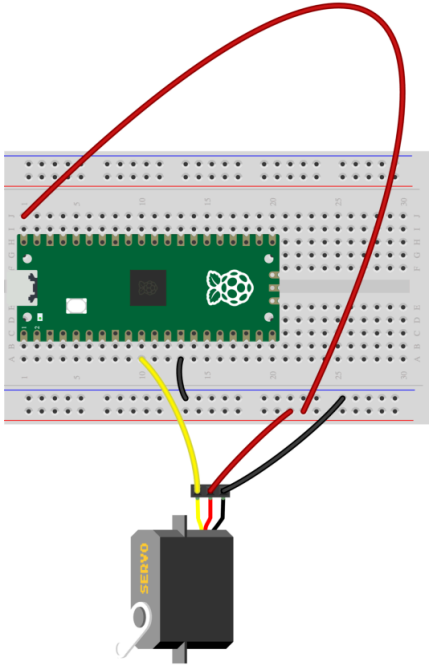
```
from picozero import Servo
from time import sleep

servo = Servo(7)
servo.mid()
sleep(2)
```

Test unloaded: the horn centres. Try min() or max() instead of mid(). Travel varies; stop if it binds or buzzes. Check lead order.

Wiring

PICO



GP7 (pin 10) → signal; **VBUS** (pin 40) → bottom + rail → servo red; **GND** (pin 13) → − rail → servo −. Small **SG90-size** servo only: USB powers it.

Stepper motor

DO | PICO | OUTPUT | MOVEMENT

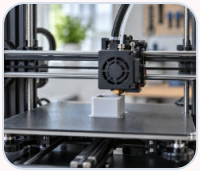
What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

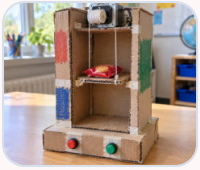
The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



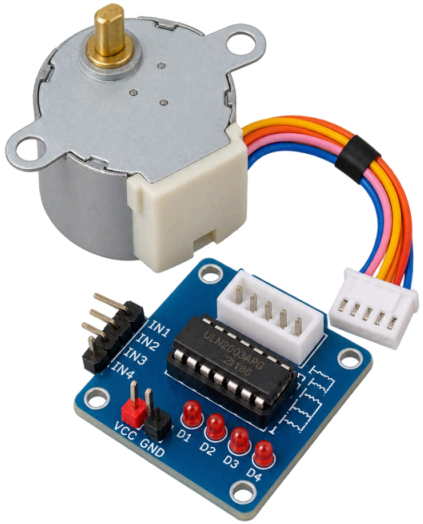
3D printer positioning

DETECT Read next print position → **DECIDE** How many steps are needed? → **DO** Move the print head



Snack lift

DETECT Snack request arrives → **DECIDE** Which floor needs it? → **DO** Step lift into place



Output shaft Moves in measured steps

Five-wire plug Connects to the driver

IN1 to IN4 Coil sequence

5V and GND Driver power

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Stepper

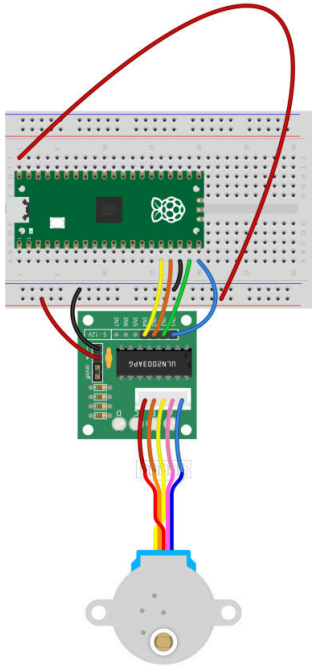
motor = Stepper((15, 14, 13, 12))

motor.step(100)
```

Call `motor.step(100)` again to keep moving. Add `direction="ccw"` to turn the other way.

Wiring

PICO



IN1-IN4 → GP **15, 14, 13, 12** (pins 20, 19, 17, 16); driver → **VBUS** (pin 40, bottom + rail) and **GND** (pin 18)